

Paper 1 - Set 09 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Quadratic functions
2	F	Polynomials, factor and remainder theorems
3	B	Trigonometric functions and equations
4	B	Angles, polygons, congruence and vectors
5	A	Functions and transformations
6	B	Ratio, proportion and percentages
7	C	Sequences and series
8	F	Integration
9	B	Exponentials and logarithms
10	C	Integration
11	F	Sequences and series
12	D	Trigonometric functions and equations
13	A	Differentiation
14	A	Probability
15	E	Exponentials and logarithms
16	B	Quadratic functions
17	A	Graphs of functions
18	E	Graphs of functions
19	D	Ratio, proportion and percentages
20	A	Straight lines

Worked solutions

Question 1 QUA-13 · Quadratic functions · spec MM1.3

Which of the following quadratics has its vertex in the second quadrant (where $x < 0$ and $y > 0$)?

Answer: D

Complete the square: A vertex $(-2, -3)$; B $(2, 1)$; C $(-1, -2)$; D $(-3, 1)$, which has $x < 0$ and $y > 0$; E $(1, 2)$. So D.

Question 2 POL-04 · Polynomials, factor and remainder theorems · spec MM1.6

Find the remainder when $x^4 + 2x^3 - 3x + 1$ is divided by $(2x - 1)$.

Answer: F

$$f\left(\frac{1}{2}\right) = \frac{1}{16} + \frac{2}{8} - \frac{3}{2} + 1 = \frac{1}{16} + \frac{4}{16} - \frac{24}{16} + \frac{16}{16} = -\frac{3}{16}.$$

Question 3 TRG-08 · Trigonometric functions and equations · spec MM4.6

Find the smallest positive solution of $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$.

Answer: B

$$x - \frac{\pi}{6} = \frac{2\pi}{3} \text{ gives } x = \frac{5\pi}{6} \text{ (the other solution } x - \frac{\pi}{6} = \frac{4\pi}{3} \text{ gives } \frac{3\pi}{2}, \text{ larger).}$$

Question 4 GEO-12 · Angles, polygons, congruence and vectors · spec M5.6

A shape is reflected in the line $y = x$ and then rotated 90° anticlockwise about the origin. The point $(3, 1)$ ends up at

Answer: B

Reflection in $y = x$: $(3, 1) \rightarrow (1, 3)$. Rotation 90° anticlockwise: $(x, y) \rightarrow (-y, x)$, so $(1, 3) \rightarrow (-3, 1)$.

Question 5 FUN-06 · Functions and transformations · spec MM8.2

The curve $y = x^2 - 6x + 5$ is reflected in the y -axis. Find the equation of the reflected curve.

Answer: A

Reflecting in the y -axis replaces x with $-x$: $y = x^2 + 6x + 5$.

Question 6 RAT-03 · Ratio, proportion and percentages · spec M3.4

£600 is divided between A, B and C in the ratio $2 : 3 : 7$. How much more does C receive than A?

Answer: B

Each part is £50; $C - A = 5 \text{ parts} = £250$.

Question 7 SEQ-02 · Sequences and series · spec MM2.3

Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

Answer: C

$$a = 12, r = -\frac{1}{3}. S_\infty = \frac{12}{1 + \frac{1}{3}} = \frac{12}{4/3} = 9.$$

Question 8 INT-02 · Integration · spec MM7.1

Find the area enclosed between the curve $y = x^2 - 4x$ and the x -axis.

Answer: F

Roots 0 and 4; the curve is below the axis between them. $\int_0^4 (x^2 - 4x) dx = \frac{64}{3} - 32 = -\frac{32}{3}$, so the area is $\frac{32}{3}$.

Question 9 LOG-12 · Exponentials and logarithms · spec MM5.1

The curve $y = a^x$ passes through (2, 12) and (4, 108). Find a .

Answer: B

$\frac{a^4}{a^2} = a^2 = \frac{108}{12} = 9$, so $a = 3$ (as $a > 0$).

Question 10 INT-03 · Integration · spec MM7.3

Given that $\int_1^5 f(x) dx = 7$ and $\int_3^5 f(x) dx = 2$, find $\int_1^3 (2f(x) - 1) dx$.

Answer: C

$\int_1^3 f = 7 - 2 = 5$. So $2 \times 5 - (3 - 1) = 8$.

Question 11 SEQ-06 · Sequences and series · spec MM2.2

The sum of the first n terms of a series is $S_n = 2n^2 + 3n$. Find the 10th term.

Answer: F

$u_{10} = S_{10} - S_9 = (200 + 30) - (162 + 27) = 230 - 189 = 41$.

Question 12 TRG-04 · Trigonometric functions and equations · spec MM4.6

Find the sum of all solutions of $\tan x = \sqrt{3}$ in the interval $0 \leq x \leq 2\pi$.

Answer: D

$x = \frac{\pi}{3}$ and $x = \frac{\pi}{3} + \pi = \frac{4\pi}{3}$. Sum $\frac{5\pi}{3}$.

Question 13 DIF-07 · Differentiation · spec MM6.3

The curve $y = ax^2 + bx$ has a stationary point at (2, -8). Find $a + b$.

Answer: A

$y' = 2ax + b = 0$ at $x = 2$: $b = -4a$. $y(2) = 4a + 2b = 4a - 8a = -4a = -8 \Rightarrow a = 2$, $b = -8$. $a + b = -6$.

Question 14 PRO-10 · Probability · spec M7.7

A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.

Answer: A

$\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} = \frac{25}{216}$.

Question 15 LOG-17 · Exponentials and logarithms · spec MM5.3

The equation $9^x - k \cdot 3^x + 9 = 0$ has exactly one real solution. Find the value of k .

Answer: E

With $y = 3^x > 0$: $y^2 - ky + 9 = 0$. Exactly one solution for x means exactly one positive root: a repeated root ($k^2 = 36$, product $9 > 0$ so the roots have the sign of k): $k = 6$ (repeated root $y = 3$, $x = 1$). $k = -6$ gives $y = -3$, no solutions. (Two distinct positive roots would give two solutions.)

Question 16 QUA-14 · Quadratic functions · spec MM1.3

The equation $x^2 - 2mx + m + 6 = 0$ has two distinct positive roots. Find the complete set of values of m .

Answer: B

Distinct real roots: $4m^2 - 4(m + 6) > 0 \Rightarrow m^2 - m - 6 > 0 \Rightarrow (m - 3)(m + 2) > 0$, so $m < -2$ or $m > 3$. Both roots positive needs sum $2m > 0$ and product $m + 6 > 0$, so $m > 0$. Combined: $m > 3$.

Question 17 GRA-13 · Graphs of functions · spec MM8.7

For how many values of k does the equation $|x^2 - 4| = k$ have exactly three real solutions?

Answer: A

The graph of $y = |x^2 - 4|$ is a W-shape with minima 0 at $x = \pm 2$ and a local maximum 4 at $x = 0$. A horizontal line gives three solutions only when it passes through the local maximum: $k = 4$. One value.

Question 18 GRA-14 · Graphs of functions · spec MM8.1

The graph of $y = \sqrt{x}$ is transformed to $y = \sqrt{4 - x}$. Which sequence of transformations achieves this?

Answer: E

$\sqrt{4 - x} = \sqrt{-(x - 4)}$: reflect $y = \sqrt{x}$ in the y-axis to get $\sqrt{-x}$, then translate 4 to the right. (Translating first then reflecting gives $\sqrt{-x - 4}$.)

Question 19 RAT-09 · Ratio, proportion and percentages · spec M3.9

p is directly proportional to $\sqrt{q}$, and q is inversely proportional to r . Which of the following is true?

Answer: D

$p = a\sqrt{q}$ and $q = \frac{b}{r}$, so $p = a\sqrt{\frac{b}{r}} = \frac{a\sqrt{b}}{\sqrt{r}}$.

Question 20 LIN-08 · Straight lines · spec MM3.1

For which value of k are the lines $kx + 3y = 2$ and $2x - (k - 1)y = 5$ perpendicular?

Answer: A

Gradients $-\frac{k}{3}$ and $\frac{2}{k-1}$. Perpendicular when the product is -1 : $\frac{-2k}{3(k-1)} = -1 \Rightarrow 2k = 3k - 3 \Rightarrow k = 3$.